

3 Germinating cereal grains, such as barley, produce the enzyme amylase.

The production of amylase is affected by gibberellin, a plant growth regulator.

A student formed the following hypothesis.

The higher the concentration of gibberellin, the greater the production of amylase by germinating cereal grains.

Plan an investigation to test this hypothesis.

(a) State two safety issues you would need to take into account.

(2)

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(b) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

(c) Devise a detailed method, including an explanation of how you would control and monitor important variables.

(10)

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(d) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(4)

[illegible]

(e) Suggest three limitations of your proposed method.

(3)

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(Total for Question 3 = 22 marks)

TOTAL FOR PAPER = 50 MARKS